

A Finite De Bruijn Certificate Improving the Upper Bound for One-Round Randomized 2-Coloring of Cycles

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September 9, 2026
Preprint draft — not peer reviewed

Abstract

Flin, Raevskaya, Stimpert, Suomela, and Yang recently reduced one-round randomized 2-coloring of directed cycles to cuts in three-dimensional De Bruijn graphs and proved

$$0.23879 \leq p^* < 0.24118.$$

Their formally verified three-parameter construction has exact monochromatic-edge probability

$$\frac{94835}{393216} = 0.2411778767903646\dots$$

We give an explicit finite coloring of the normal De Bruijn graph $\text{DB}_{\text{normal}}(497)$ with exactly 14,714,994,317 monochromatic edges out of 61,013,446,081 directed edges. Hence

$$p^* \leq \frac{14,714,994,317}{61,013,446,081} = 0.2411762531404098\dots < \frac{94835}{393216}.$$

The coloring is the midpoint discretization of the published three-parameter rule, followed by 4750 explicitly listed vertex flips. The strict improvement is certified by exact integer arithmetic; in particular, the relevant cross-product difference is $-38,953,738,163$. We also show that no local flips are needed merely to obtain a strict improvement: the unmodified midpoint discretization at $n = 281$ already beats the published exact value. The contribution is a finite, machine-checkable upper-bound certificate within the De Bruijn framework of Flin et al.; we do not claim a new general framework, a lower bound, or optimality of the displayed finite cuts.

Keywords. distributed graph algorithms; randomized local algorithms; cycle coloring; De Bruijn graphs; maximum cut; finite certificates.

1 Introduction

Consider a directed cycle whose nodes run a randomized distributed algorithm for one communication round and then output one of two colors. A proper 2-coloring cannot always be produced, so the objective is to minimize the expected fraction of monochromatic edges. In the standard continuous-randomness formulation, a one-round algorithm is a measurable function

$$f : [0, 1]^3 \rightarrow \{0, 1\},$$

where the three arguments are the independent random values held by a node's predecessor, the node itself, and its successor. If A, B, C, D are i.i.d. $\text{Unif}[0, 1]$, define

$$p(f) := \Pr[f(A, B, C) = f(B, C, D)], \quad p^* := \inf_f p(f).$$

Flin et al. [2, 1] developed a systematic finite reformulation of this problem using normal and distinct three-dimensional De Bruijn graphs. Their main result is

$$0.23879 \leq p^* < 0.24118.$$

For the upper bound, they discovered a three-parameter recursive cutoff rule and formally verified its exact value in Lean 4. In the public formalization at the pinned commit cited in the bibliography, the file `Distributed2Coloring/UpperBound/Recursive3Param.lean` defines

$$t = \frac{5}{8}, \quad t_1 = \frac{3}{8}, \quad t_2 = \frac{17}{32},$$

and the file `Distributed2Coloring/UpperBound/Recursive3Param/Final.lean` proves [3]

$$p_0 := \frac{94835}{393216} = 0.2411778767903646 \dots$$

The repository's `Distributed2Coloring/MainResults.lean` packages the resulting public statement $p^* < 0.24118$; it does not contain the exact rational equality, which is proved in the `Recursive3Param` files just cited.

This note asks a deliberately narrow question: can the same De Bruijn framework provide a finite coloring whose exact ratio is strictly smaller than p_0 ? The answer is yes.

Main result. We construct an explicit coloring h_{497} of $\text{DB}_{\text{normal}}(497)$ with

$$M_{497} = 14,714,994,317$$

monochromatic edges among

$$497^4 = 61,013,446,081$$

directed edges. By the De Bruijn upper-bound lemma of Flin et al., this gives

$$p^* \leq \frac{M_{497}}{497^4} = 0.2411762531404098 \dots$$

The comparison with p_0 is exact:

$$M_{497} \cdot 393216 - 94835 \cdot 497^4 = -38,953,738,163 < 0.$$

Thus the new bound is not a floating-point artifact.

A simpler strict improvement. The local-refinement step is not essential for the qualitative statement. The midpoint discretization of the published three-parameter rule on $\text{DB}_{\text{normal}}(281)$, with no vertex flips, has exactly 1,503,704,985 monochromatic edges among $281^4 = 6,234,839,521$ edges, and

$$1,503,704,985 \cdot 393216 - 94835 \cdot 281^4 = -146,592,275 < 0.$$

Hence even this short deterministic finite discretization already yields a strictly smaller certified upper bound than p_0 .

Scope. The novelty claimed here is only the new finite upper-bound certificate and its exact verification. The reduction from one-round randomized cycle coloring to De Bruijn cuts, the three-parameter base rule, and the idea of searching for good finite cuts all come from Flin et al. [2]. We do not claim that h_{497} is an optimal coloring of $\text{DB}_{\text{normal}}(497)$, and we do not improve the lower bound 0.23879.

2 De Bruijn formulation

For an integer $n \geq 2$, write $[n] = \{0, 1, \dots, n-1\}$. The normal three-dimensional De Bruijn graph is

$$\text{DB}_{\text{normal}}(n) = (V_n, E_n),$$

where

$$V_n = [n]^3$$

and

$$E_n = \{((a, b, c), (b, c, d)) : a, b, c, d \in [n]\}.$$

Therefore

$$|V_n| = n^3, \quad |E_n| = n^4.$$

For a coloring $h : V_n \rightarrow \{0, 1\}$, let

$$M(h) := \#\{(a, b, c, d) \in [n]^4 : h(a, b, c) = h(b, c, d)\}$$

be its number of monochromatic directed edges, and put

$$\rho(h) := \frac{M(h)}{n^4}.$$

Let $p_{\text{normal}}(n)$ be the minimum of $\rho(h)$ over all such colorings.

We use the following consequence of Lemma 4.1 of Flin et al. [2].

Lemma 1 (Finite De Bruijn certificate). *For every $n \geq 2$ and every coloring $h : V_n \rightarrow \{0, 1\}$,*

$$p^* \leq p_{\text{normal}}(n) \leq \rho(h).$$

Proof. The first inequality is exactly the normal-De-Bruijn upper-bound lemma of Flin et al. The second follows from the definition of $p_{\text{normal}}(n)$ as the minimum monochromatic-edge fraction among all colorings of $\text{DB}_{\text{normal}}(n)$. $\square$

Thus a new upper bound on p^* can be proved by exhibiting any explicit finite coloring and counting its monochromatic edges exactly.

3 Base coloring

We start from the three-parameter recursive cutoff rule of Flin et al. [3]. The parameter definitions below are a direct transcription of `Distributed2Coloring/UpperBound/Recursive3Param.lean` at the pinned commit cited in the bibliography. This section states the rule explicitly so that the finite certificate does not depend on an informal geometric description. Set

$$t_1 = \frac{3}{8}, \quad t_2 = \frac{17}{32}, \quad t = \frac{5}{8}.$$

Define the base cutoff $z_{\text{base}} : [0, 1]^2 \rightarrow [0, 1]$ by

$$z_{\text{base}}(x, y) = \begin{cases} 1, & y \geq t \text{ and } x < t, \\ t, & y \geq t \text{ and } x \geq t, \\ 0, & y < t \text{ and } x \geq t, \\ t, & y < t, x < t, x \leq y, \\ y, & y < t, x < t, x > y. \end{cases}$$

Inside the square $[t_1, t]^2$, replace this by

$$z_0(x, y) = \begin{cases} t_1, & y < t_2, x \geq t_2, \\ t_2, & y < t_2, x < t_2, x \leq y, \\ y, & y < t_2, x < t_2, x > y, \\ t, & y \geq t_2, x < t_2, \\ t_2, & y \geq t_2, x \geq t_2, \end{cases}$$

and outside $[t_1, t]^2$ set $z_0 = z_{\text{base}}$. Finally define

$$f_0(x, y, z) := \mathbf{1}\{z < z_0(x, y)\}.$$

This is the same rational-parameter rule formalized by Flin et al. [3].

For a finite size n , use midpoint representatives

$$x_i := \frac{2i + 1}{2n}, \quad i \in [n],$$

and define the midpoint discretization

$$h_0^{(n)}(i, j, k) := f_0(x_i, x_j, x_k).$$

All comparisons against t_1, t_2, t can be implemented using integer cross-products, so no floating-point arithmetic is needed.

Proposition 2 (Strict improvement without local flips). *For $n = 281$, the midpoint coloring $h_0^{(281)}$ has*

$$M(h_0^{(281)}) = 1,503,704,985.$$

Consequently,

$$p^* \leq \frac{1,503,704,985}{6,234,839,521} = 0.2411778169967753 \dots < \frac{94835}{393216}.$$

Proof. The exact checker evaluates the midpoint rule and counts all directed edges by the aggregation identity proved in Section 5. It returns

$$M = 1,503,704,985, \quad |E| = 281^4 = 6,234,839,521.$$

The strict comparison is certified by

$$1,503,704,985 \cdot 393216 - 94835 \cdot 6,234,839,521 = -146,592,275 < 0.$$

Apply the finite De Bruijn certificate lemma. □

4 Refined certificate at $n = 497$

The strongest certificate reported here uses the same midpoint base coloring at $n = 497$ and then toggles a finite set of vertices.

Let

$$F \subseteq [497]^3, \quad |F| = 4750,$$

be the explicit list supplied in the machine-readable certificate `n497_certificate.json`. Define

$$h_{497}(v) := h_0^{(497)}(v) \oplus \mathbf{1}\{v \in F\}.$$

The full list is deliberately kept outside the manuscript; the theorem depends only on that fixed finite file and the exact checker.

Search used for the reported refinement. For reproducibility, the artifact package also contains the deterministic search that generated F . Let $W = \{218, 219, \dots, 279\}$. Starting from $h_0^{(497)}$, for every candidate $v \in W^3$ let $\Delta M(v)$ be the exact change in the monochromatic-edge count produced by toggling v . At each step, the search chooses a vertex with minimum negative $\Delta M(v)$, breaking ties lexicographically by (a, b, c) , applies that toggle, and repeats. It stops when no negative change remains. All changes are computed with integer counts. This procedure terminates after exactly 4750 flips; the cumulative change is $-77,600$, and the best remaining change is 0. The resulting ordered flip list reproduces the supplied certificate exactly. This search description is included for experimental reproducibility and is not needed for the validity of the fixed finite certificate.

For context, before applying the flips, the midpoint coloring already has

$$M(h_0^{(497)}) = 14,715,071,917,$$

corresponding to

$$0.2411775249912064\dots,$$

which is itself below p_0 . The 4750 flips reduce the monochromatic-edge count by exactly 77,600.

Theorem 3 (Improved finite upper bound). *The coloring h_{497} has exactly*

$$M(h_{497}) = 14,714,994,317$$

monochromatic directed edges. Therefore

$$p^* \leq \frac{14,714,994,317}{61,013,446,081} = 0.2411762531404098\dots$$

and, in particular,

$$\frac{14,714,994,317}{61,013,446,081} < \frac{94835}{393216}.$$

Proof. The exact checker evaluates the fixed coloring h_{497} and returns

$$\begin{aligned} |V_{497}| &= 497^3 = 122,763,473, \\ M(h_{497}) &= 14,714,994,317, \\ |E_{497}| - M(h_{497}) &= 46,298,451,764, \\ |E_{497}| &= 497^4 = 61,013,446,081. \end{aligned}$$

Hence the monochromatic-edge fraction is the displayed rational number. The exact cross-product comparison with the published formally verified value is

$$14,714,994,317 \cdot 393216 - 94,835 \cdot 61,013,446,081 = -38,953,738,163 < 0.$$

The finite De Bruijn certificate lemma now gives the claimed upper bound on p^* . $\square$

The numerical improvement over p_0 is

$$\frac{94835}{393216} - \frac{14,714,994,317}{61,013,446,081} = \frac{38,953,738,163}{23,991,463,214,186,496} \approx 1.6236499548 \times 10^{-6}.$$

It is small, but exact.

5 Exact verification

A literal traversal of all n^4 directed edges is unnecessary. For each ordered pair $(b, c) \in [n]^2$, define

$$L_1(b, c) := \sum_{a \in [n]} h(a, b, c), \quad R_1(b, c) := \sum_{d \in [n]} h(b, c, d).$$

Among the n^2 directed edges of the form

$$(a, b, c) \rightarrow (b, c, d)$$

with fixed (b, c) , the number of monochromatic edges is

$$L_1(b, c)R_1(b, c) + (n - L_1(b, c))(n - R_1(b, c)).$$

Similarly, the cut-edge count is

$$L_1(b, c)(n - R_1(b, c)) + (n - L_1(b, c))R_1(b, c).$$

Summing over (b, c) yields the exact total in $O(n^3)$ arithmetic operations.

Lemma 4 (Aggregation identity). *For every coloring $h : [n]^3 \rightarrow \{0, 1\}$,*

$$M(h) = \sum_{b, c \in [n]} [L_1(b, c)R_1(b, c) + (n - L_1(b, c))(n - R_1(b, c))].$$

Proof. Fix (b, c) . A directed edge $(a, b, c) \rightarrow (b, c, d)$ is monochromatic with color 1 exactly when both endpoints have color 1, giving $L_1 R_1$ choices. It is monochromatic with color 0 exactly when both endpoints have color 0, giving $(n - L_1)(n - R_1)$ choices. These cases are disjoint and exhaustive among monochromatic edges with the fixed middle pair (b, c) . Summing over all n^2 middle pairs counts every directed edge exactly once. $\square$

The supplied C++ checker uses unsigned 64-bit integers for the edge counts and 128-bit integers for rational cross-products. At $n = 497$, all raw counts are well below 2^{64} . The final run verifies both

$$M + \text{cut} = n^4$$

and the exact target count before returning success.

Table 1: Exact values relevant to the upper-bound improvement. The last column is $M \cdot 393216 - 94835 \cdot n^4$; a negative value certifies a strict improvement over $94835/393216$.

Certificate	n	Monochromatic edges M	Fraction	Cross product
Midpoint base	281	1,503,704,985	0.2411778169967753	-146,592,275
Midpoint base	497	14,715,071,917	0.2411775249912064	-8,440,176,563
Refined	497	14,714,994,317	0.2411762531404098	-38,953,738,163

6 Reproducibility and certificate boundary

Artifact availability in this proofreading draft. The certificate files listed below are distributed with this manuscript in `Yonezu_Paper11_Proofreading_Bundle_v0.5.zip`. A permanent public GitHub repository and Zenodo software DOI have not yet been assigned; those identifiers are a required prepublication item and will replace this draft-local locator before public release. No public URL or DOI is fabricated in the present draft.

The certificate package contains:

1. `n497_certificate.json`, containing the base-rule metadata and the explicit list of 4750 flipped vertices;
2. `reproduce_n497_search.cpp`, the standalone deterministic local-search reproducer that starts from the midpoint base coloring and regenerates the ordered flip list without reading the certificate;
3. `SEARCH_REPRODUCTION.md`, documenting the exact one-flip objective change, tie-breaking rule, stopping rule, and update scheme;
4. `check_n497_embedded_certificate.cpp`, a standalone exact checker with the flip list embedded in source form;
5. `check_n281_base.cpp`, a standalone checker for the simpler no-flip certificate;
6. `reproduce_all.sh` and `reproduce_all.ps1`, which compile the reproducer and checkers, regenerate the $n = 497$ certificate, compare it byte-for-byte with the canonical JSON, and rerun both exact checkers;
7. logs recording the exact integer outputs of the search and checkers.

For the fixed files used in this draft, the SHA-256 values are

```
n497_certificate.json
41e62c73421e9ca33b097ecedda6d31b
398ed8189d74444258c8015e197f193c
reproduce_n497_search.cpp
4bb0bfe5467deb76a5e2861b0c13c2f
8a95de6e612ecd7d11ce5d2bbbfa7a3db
check_n497_embedded_certificate.cpp
06280be157851220138e8fec38d2852ac
da1b59543a0838040d57a3e171cf582
```

Each two-line hexadecimal value is to be concatenated without whitespace.

The theorem itself still requires only reconstructing the fixed finite coloring, evaluating the aggregation identity, and checking one integer inequality. The additional search reproducer closes the stronger experimental reproducibility loop: base coloring $\rightarrow$ deterministic local search $\rightarrow$ regenerated certificate $\rightarrow$ exact checker. The reproduced JSON is byte-identical to the canonical certificate. No stochastic optimization, floating-point tolerance, or solver output is involved in either the search reproduction or the proof-certificate verdict.

7 Discussion and limitations

The result improves the published exact value of the three-parameter construction of Flin et al., but the improvement is numerically modest. It narrows the interval between the published lower bound 0.23879 and the upper bound by about 0.068% relative to the previous interval width.

More importantly, the result should not be interpreted as evidence that the finite coloring is close to optimal. We have not computed $p_{\text{normal}}(497)$ exactly, nor do we provide a lower bound on the maximum cut of $\text{DB}_{\text{normal}}(497)$. Further finite local search, stronger combinatorial optimization, or a new structured continuous rule may lower the upper bound again.

The $n = 281$ proposition also highlights a discretization effect: the finite midpoint evaluation of the existing continuous rule can lie slightly below its continuous exact value. This is fully legitimate in the De Bruijn framework because each finite coloring independently induces a valid one-round randomized algorithm through the discretization lemma. The $n = 497$ local refinement then exploits the additional freedom of arbitrary finite colorings beyond direct samples of the continuous surface.

8 Conclusion

We have exhibited a finite, exactly checkable De Bruijn coloring giving

$$p^* \leq 0.2411762531404098\dots,$$

strictly improving the formally verified value $94835/393216 = 0.2411778767903646\dots$ associated with the previously published three-parameter rule. The theorem follows from an explicit 497-symbol coloring, exact integer edge counting, and the normal De Bruijn upper-bound lemma of Flin et al. A simpler 281-symbol midpoint discretization already suffices for a strict improvement, while the refined 497-symbol certificate yields the stronger numerical bound reported here.

References

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